



# Hydrosight: Geospatial Artificial Intelligence Platform



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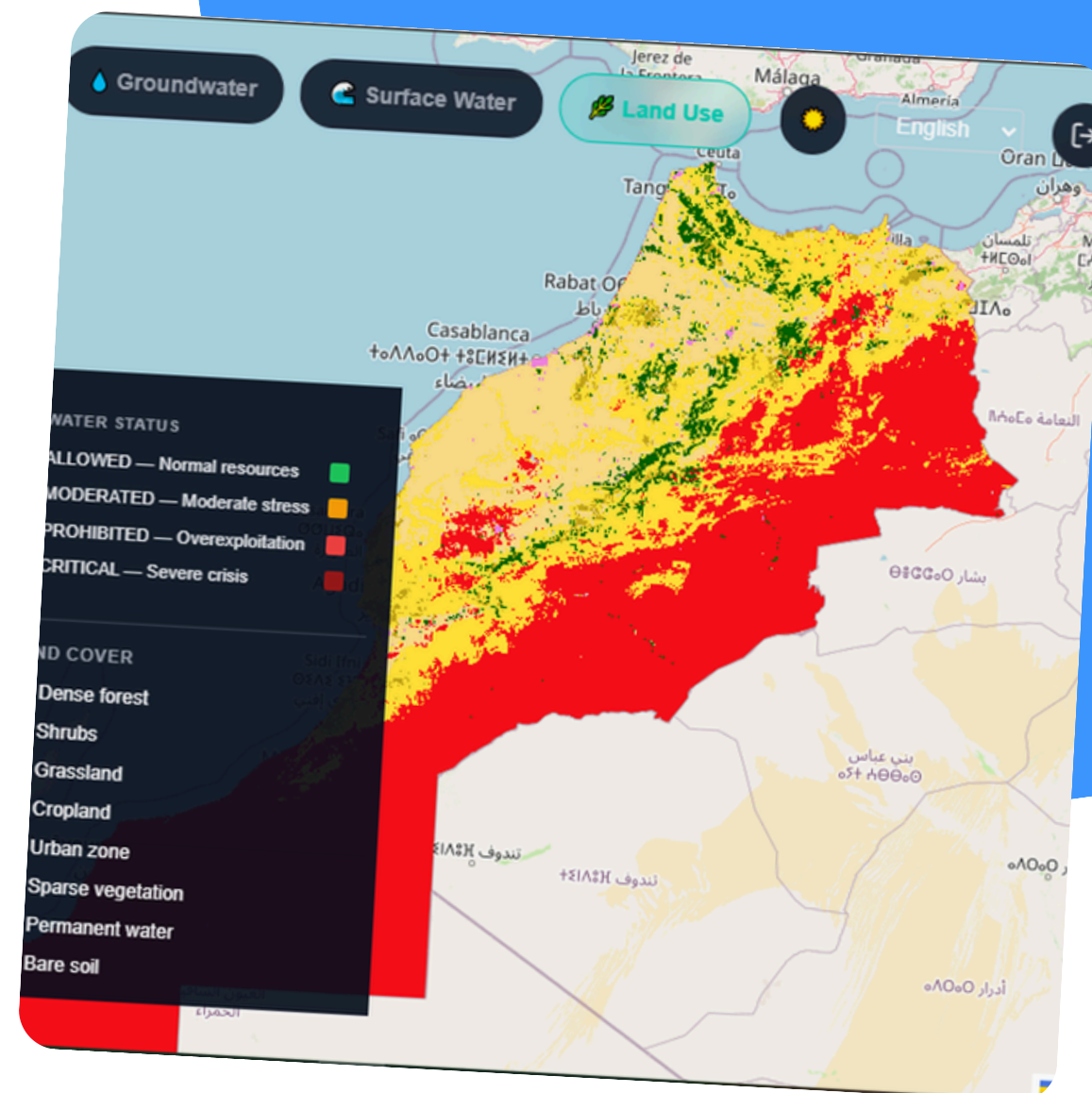
**HydroSight**  
GEOAI WATER INTELLIGENCE

1

# Contexte Général



# Problématique



Les ressources en eau diminuent, les décisions doivent s'accélérer

# **Solution proposée — HydroSight**

**Plateforme web d'analyse des ressources hydriques par satellite  
Powered by Google Earth Engine  
5 sources de données : GRACE, CHIRPS, MODIS, Sentinel-2,  
Sentinel-1**

# Objectifs

Développer une plateforme web permettant :

- la sélection d'une zone géographique ;
- la sélection d'une période d'étude ;
- l'analyse des ressources hydriques ;
- le calcul d'un score de stress hydrique.



# Analyse des besoins



# Acteurs

## > Utilisateur Invité

L'utilisateur invité peut accéder à une démonstration limitée de la plateforme sans authentication.

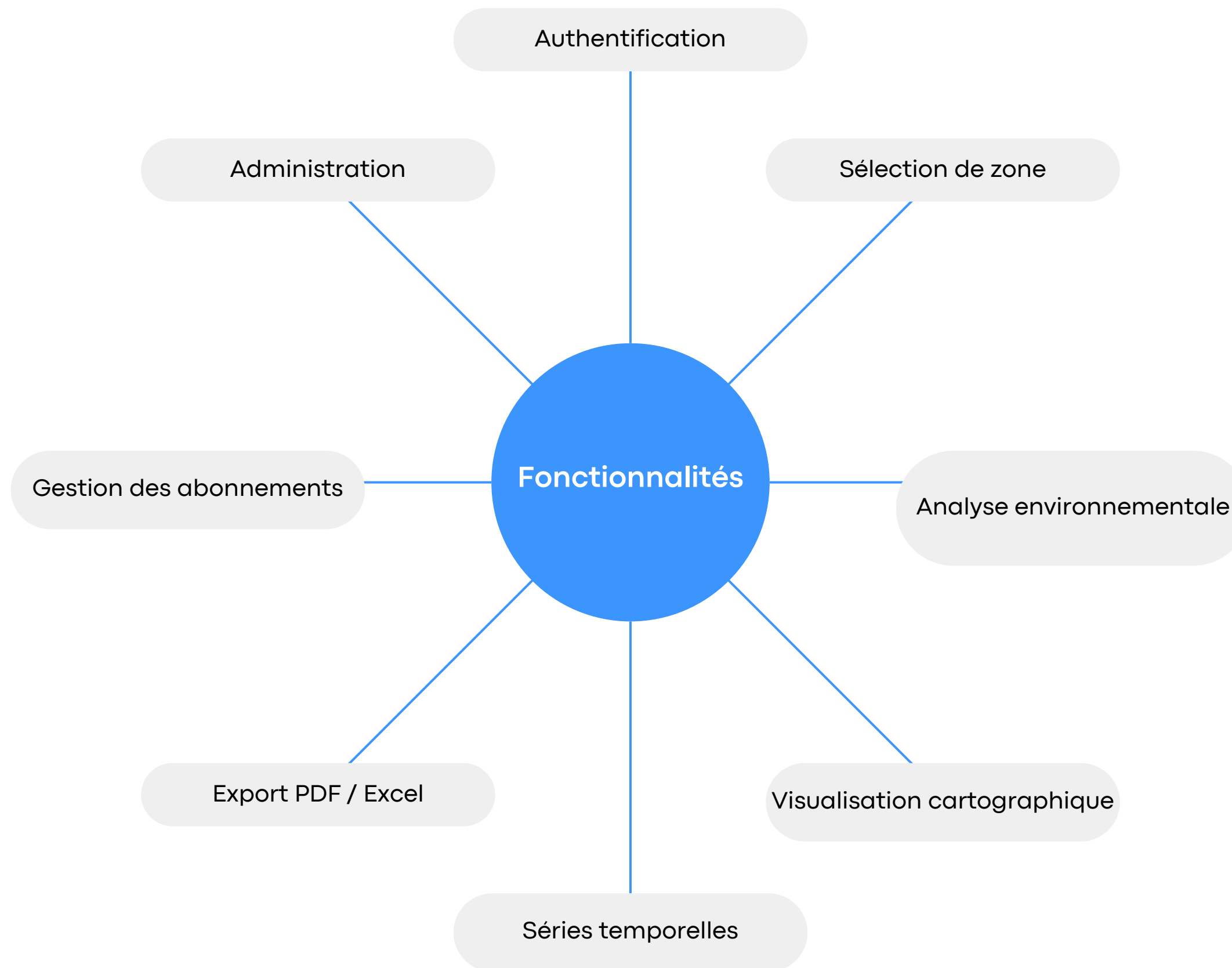
## > Utilisateur Enregistré

L'utilisateur enregistré dispose d'un compte personnel lui permettant d'accéder à l'ensemble des fonctionnalités associées à son abonnement,

## > Administrateur

L'administrateur assure la gestion et la supervision de la plateforme. Il peut gérer les utilisateurs, les contrats B2B, les abonnements

# Besoins fonctionnels



# Tableau récapitulatif des fonctionnalités par plan

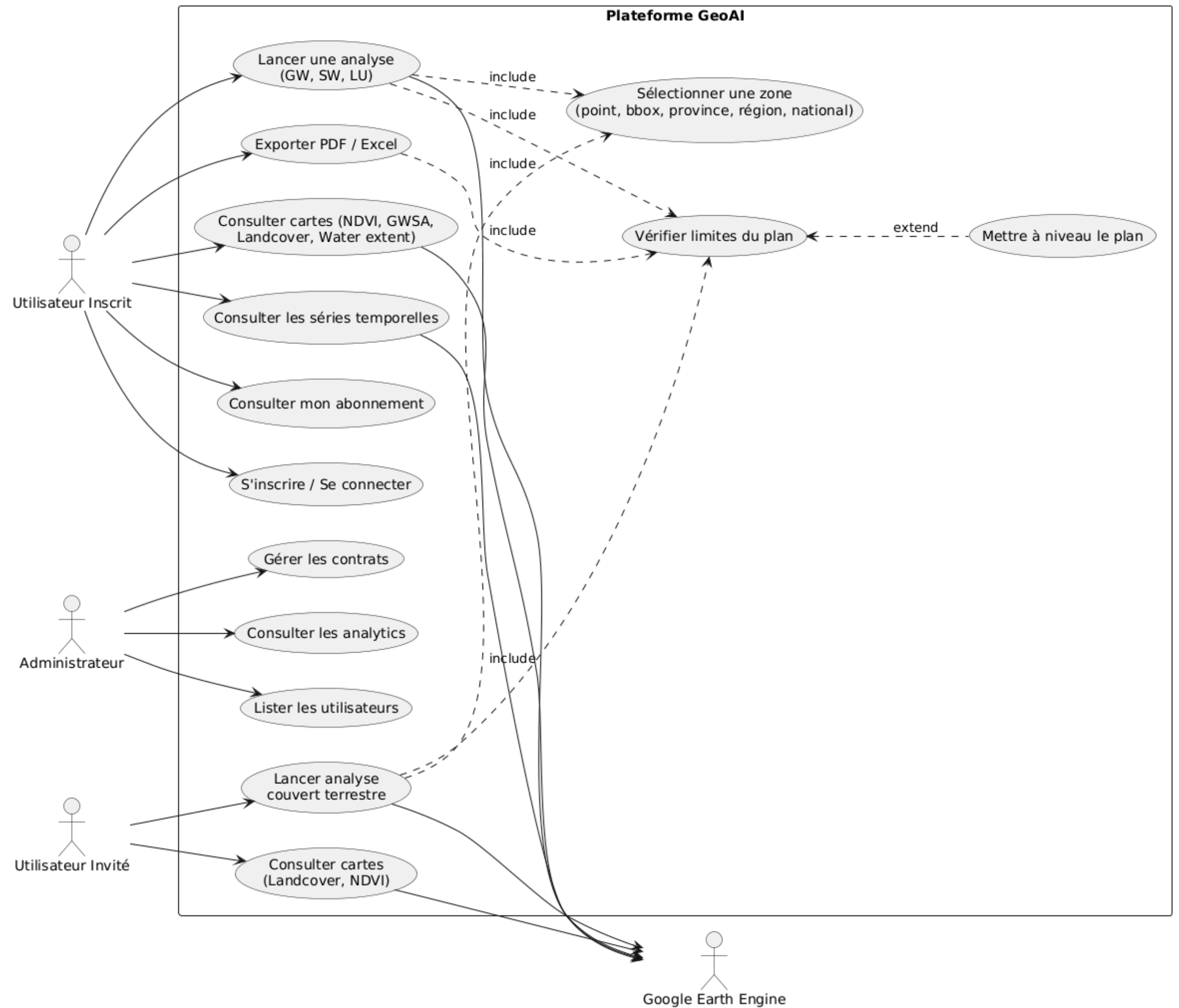
| Fonctionnalité           | Guest         | Free    | Pro      | Premium   | Contrat   |
|--------------------------|---------------|---------|----------|-----------|-----------|
| Module Couvert terrestre | ✓             | ✓       | ✓        | ✓         | ✓         |
| Module Eaux souterraines | —             | —       | ✓        | ✓         | ✓         |
| Module Eaux de surface   | —             | —       | ✓        | ✓         | ✓         |
| Mode Point               | ✓             | ✓       | ✓        | ✓         | ✓         |
| Mode Bounding Box        | —             | —       | ✓        | ✓         | ✓         |
| Mode Province            | ✓             | —       | ✓        | ✓         | ✓         |
| Mode Région              | ✓             | —       | —        | ✓         | ✓         |
| Mode National            | —             | —       | —        | ✓         | ✓         |
| Période max              | 3 mois        | 6 mois  | 36 mois  | Illimitée | Illimitée |
| Quota                    | Illimité/jour | 10/mois | 100/mois | 500/mois  | Illimité  |
| Export PDF               | —             | —       | ✓        | ✓         | ✓         |
| Export Excel             | —             | —       | ✓        | ✓         | ✓         |



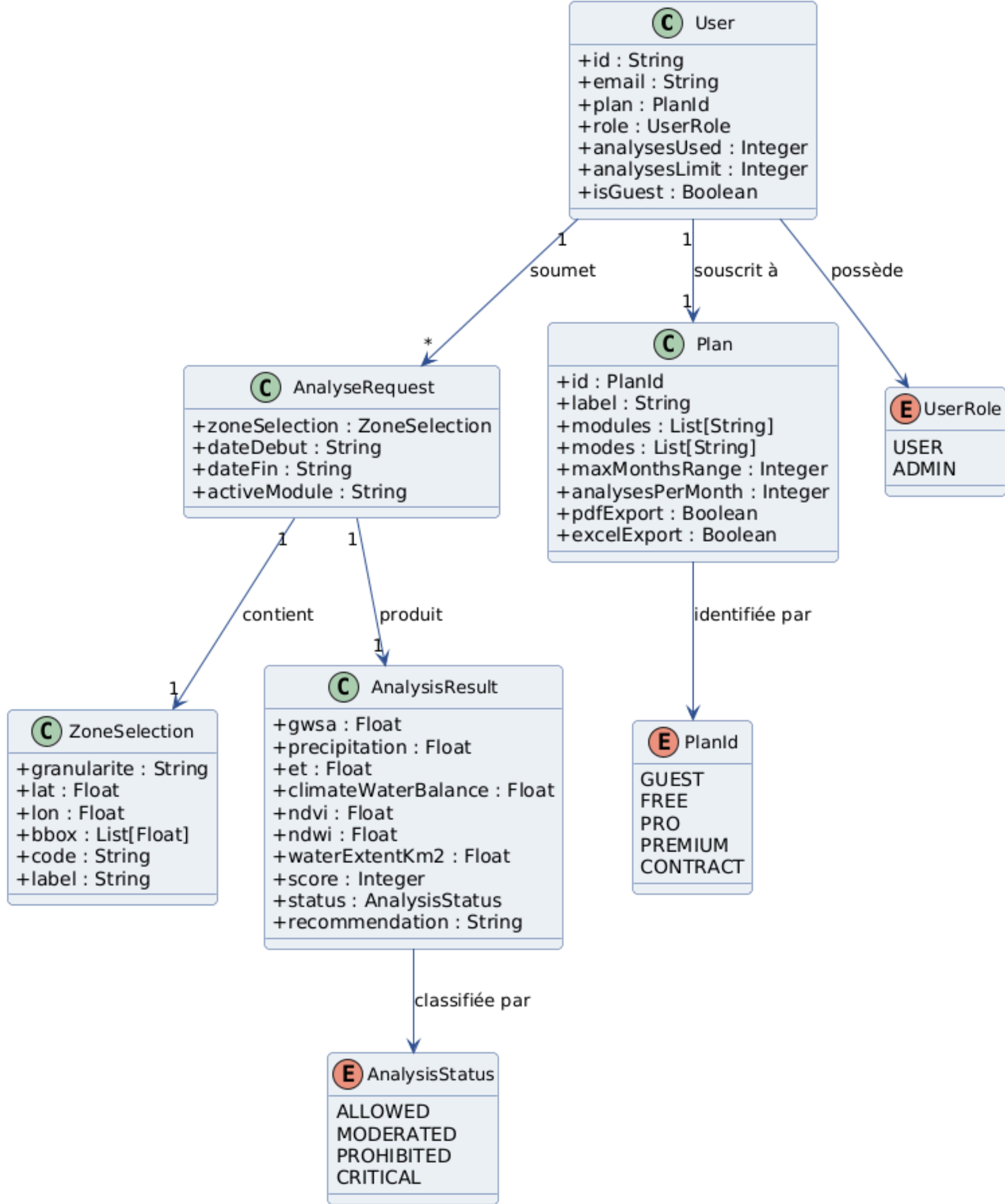
# Conception



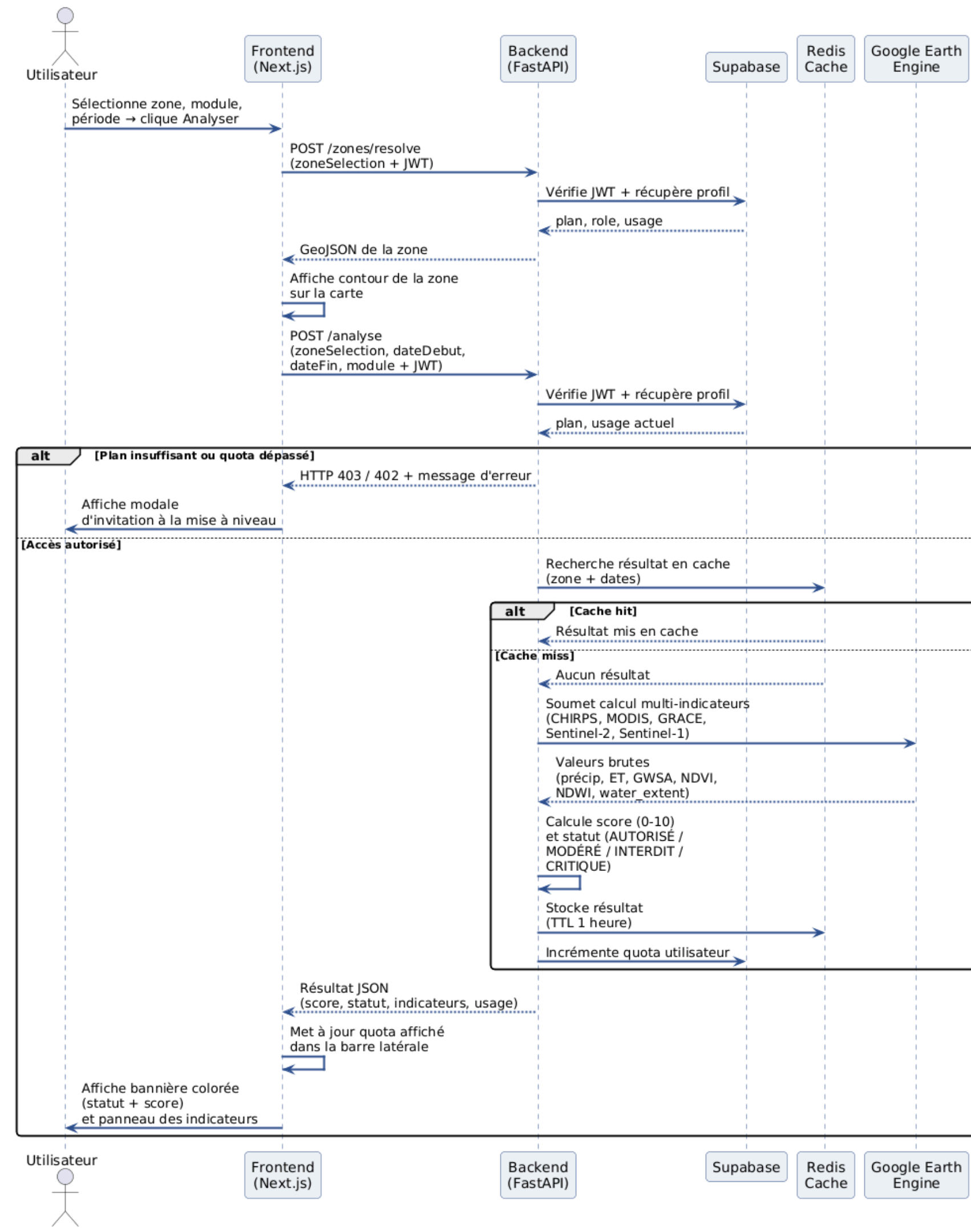
# Cas d'utilisation



# Diagramme de classes



# Diagramme de séquence





Réalisation



## Outils de réalisation

### > Intelligence Géospatiale & Traitement de Données



Google Earth Engine

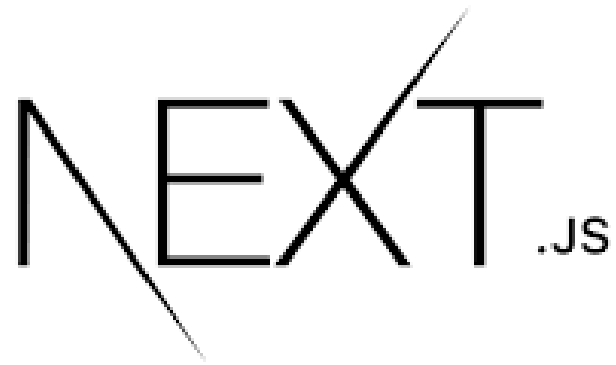
# Outils de réalisation

## > Backend

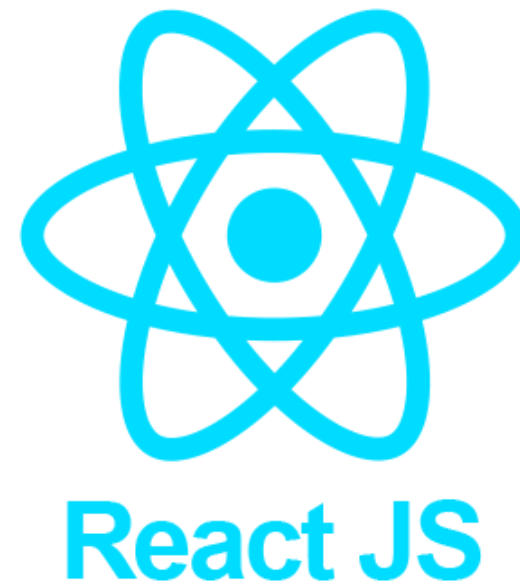


# Outils de réalisation

## > Frontend

The logo for Next.js, featuring the word "NEXT" in a large, black, sans-serif font, with ".JS" in a smaller font to the right. A thin black diagonal line crosses through the "X".

NEXT.JS

The logo for React JS, featuring a stylized blue atom with three elliptical orbits and a central blue circle. Below the atom, the text "React JS" is written in a blue, sans-serif font.

React JS

The logo for Tailwind CSS, featuring a stylized blue wave icon above the text "tailwindcss" in a black, sans-serif font.

tailwindcss

The logo for React Leaflet, featuring a green leaf shape with a white stylized atom icon inside. Below the leaf, the text "React Leaflet" is written in a black, sans-serif font.

React Leaflet

The logo for Recharts, featuring the word "Recharts" in a teal, sans-serif font.

Recharts

The logo for next-intl, featuring a blue globe icon with a grid pattern to the left of the text "next-intl" in a black, sans-serif font.

next-intl

## Outils de réalisation

### > Base de Données & Authentification



## Outils de réalisation

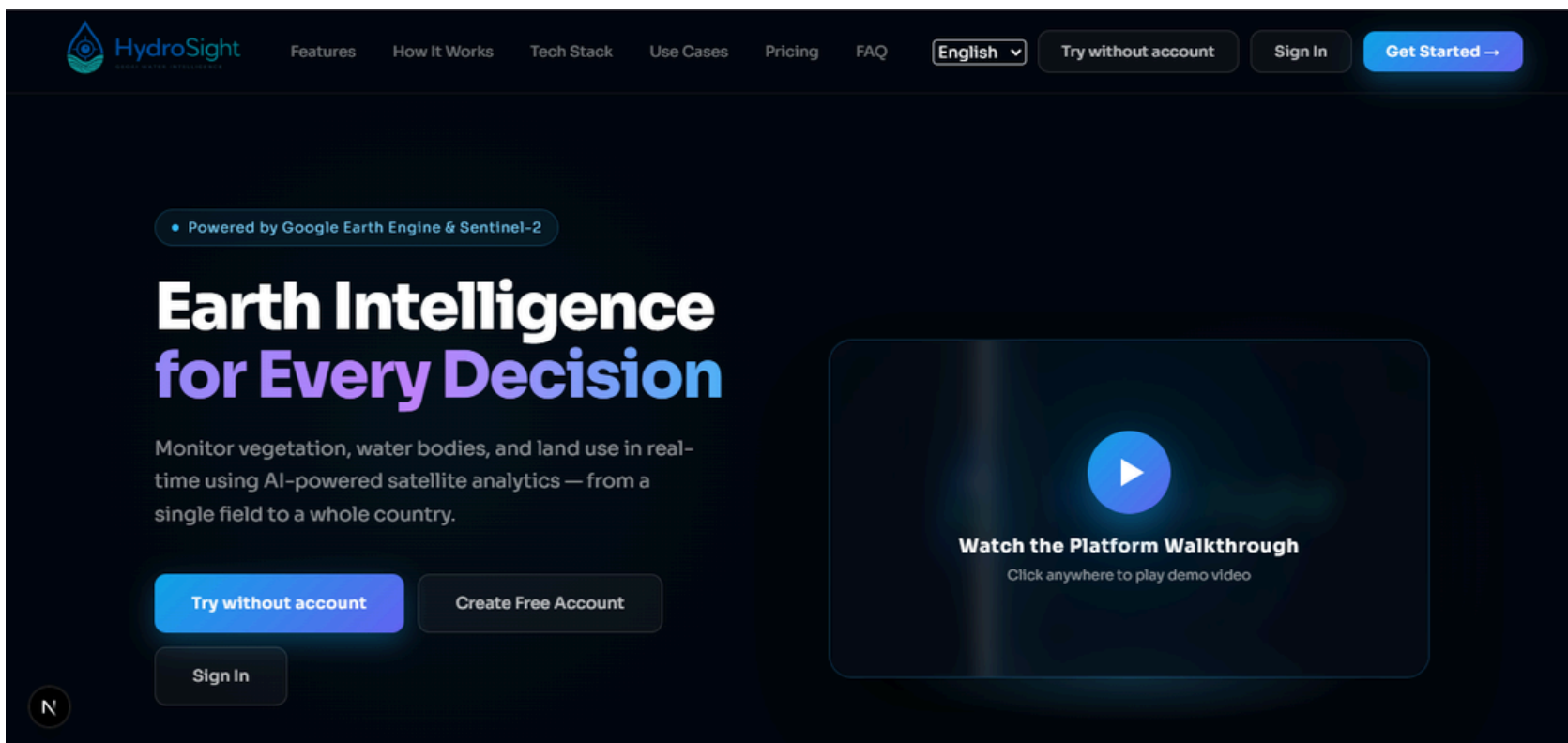
### > Performance & Optimisation



redis

# Description de la plateforme

## > Landing page



HydroSight

Features How It Works Tech Stack Use Cases Pricing FAQ English Try without account Sign In Get Started →

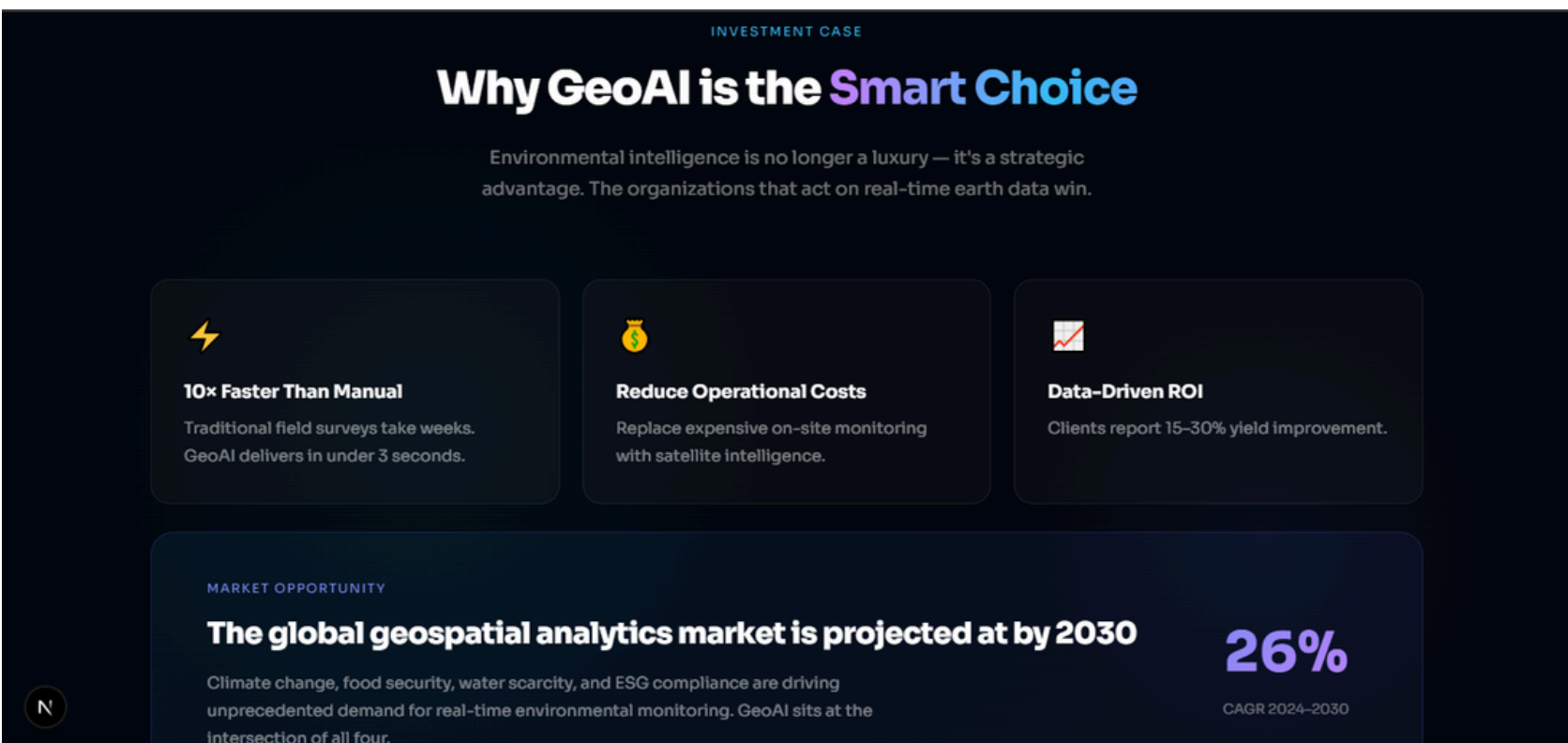
Powered by Google Earth Engine & Sentinel-2

## Earth Intelligence for Every Decision

Monitor vegetation, water bodies, and land use in real-time using AI-powered satellite analytics — from a single field to a whole country.

Try without account Create Free Account Sign In

Watch the Platform Walkthrough  
Click anywhere to play demo video



### INVESTMENT CASE

## Why GeoAI is the Smart Choice

Environmental intelligence is no longer a luxury — it's a strategic advantage. The organizations that act on real-time earth data win.

- 10× Faster Than Manual**  
Traditional field surveys take weeks. GeoAI delivers in under 3 seconds.
- Reduce Operational Costs**  
Replace expensive on-site monitoring with satellite intelligence.
- Data-Driven ROI**  
Clients report 15–30% yield improvement.

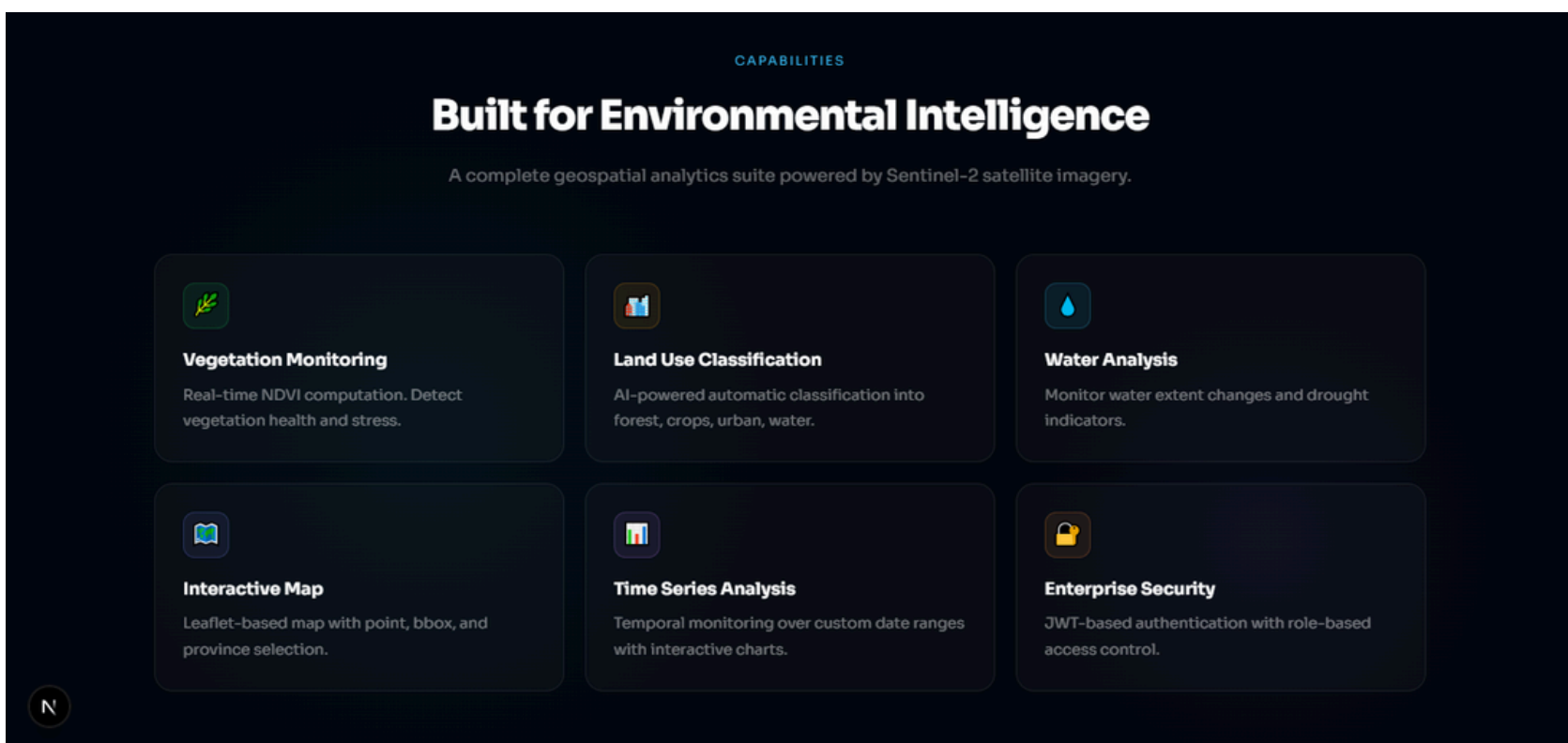
### MARKET OPPORTUNITY

## The global geospatial analytics market is projected at by 2030

Climate change, food security, water scarcity, and ESG compliance are driving unprecedented demand for real-time environmental monitoring. GeoAI sits at the intersection of all four.

# 26%

CAGR 2024–2030

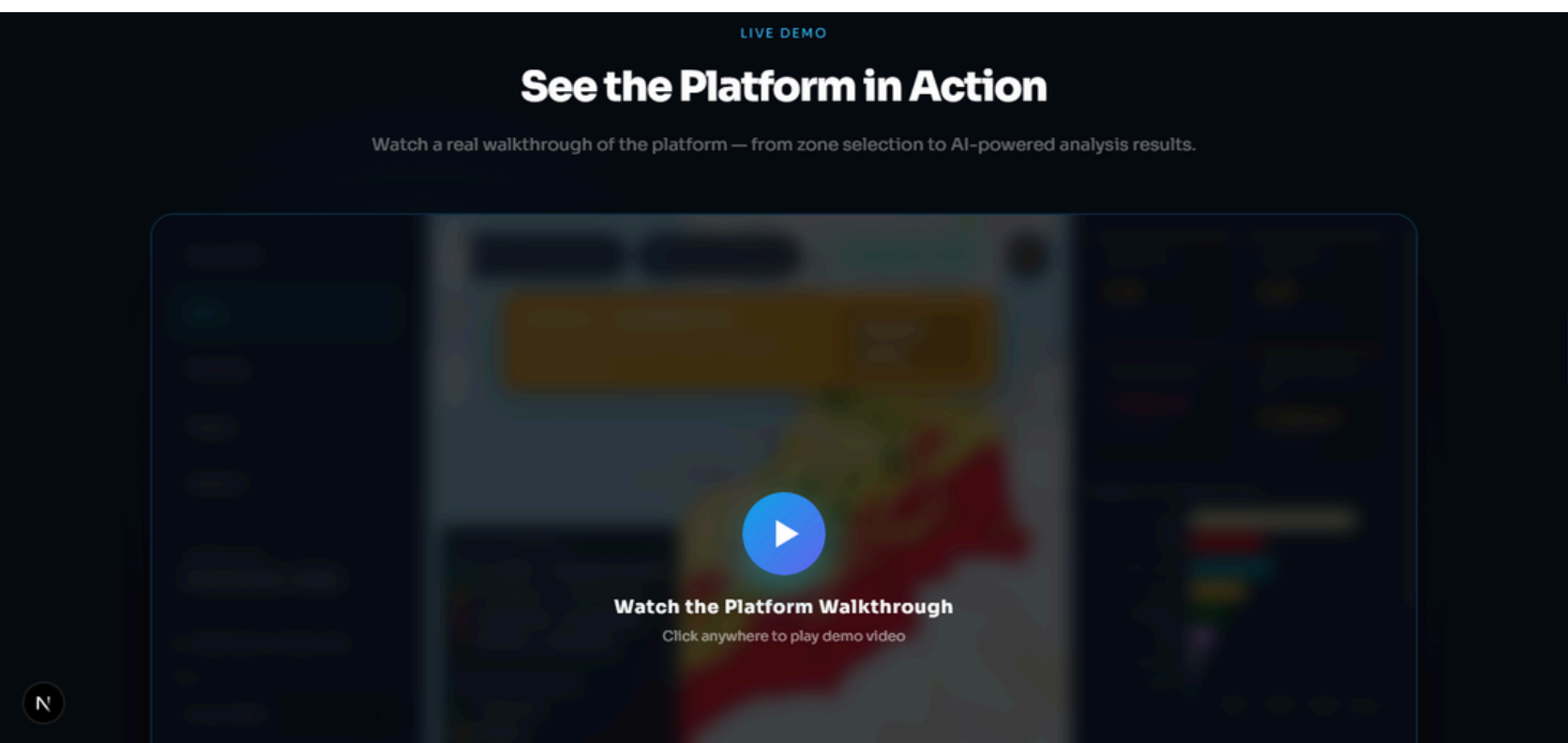


### CAPABILITIES

## Built for Environmental Intelligence

A complete geospatial analytics suite powered by Sentinel-2 satellite imagery.

- Vegetation Monitoring**  
Real-time NDVI computation. Detect vegetation health and stress.
- Land Use Classification**  
AI-powered automatic classification into forest, crops, urban, water.
- Water Analysis**  
Monitor water extent changes and drought indicators.
- Interactive Map**  
Leaflet-based map with point, bbox, and province selection.
- Time Series Analysis**  
Temporal monitoring over custom date ranges with interactive charts.
- Enterprise Security**  
JWT-based authentication with role-based access control.



### LIVE DEMO

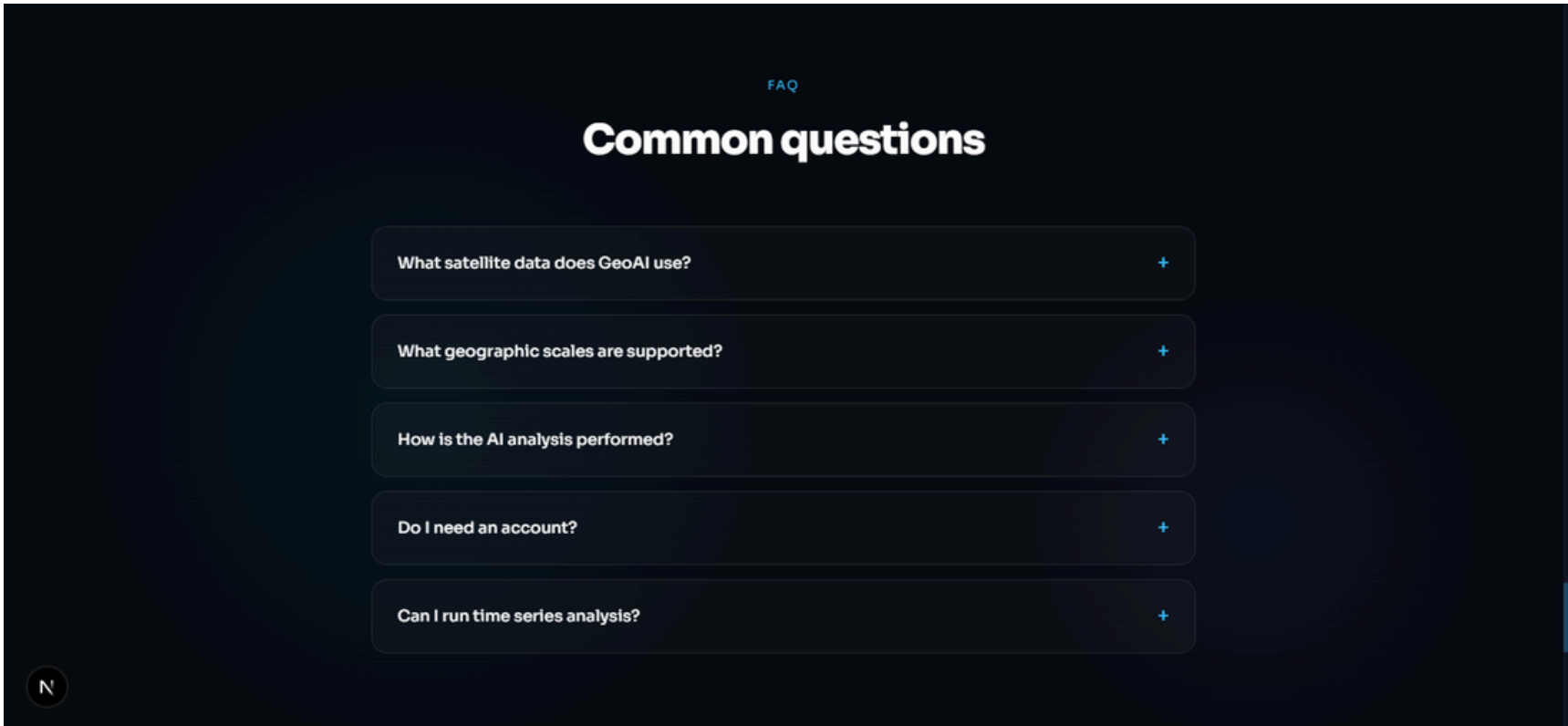
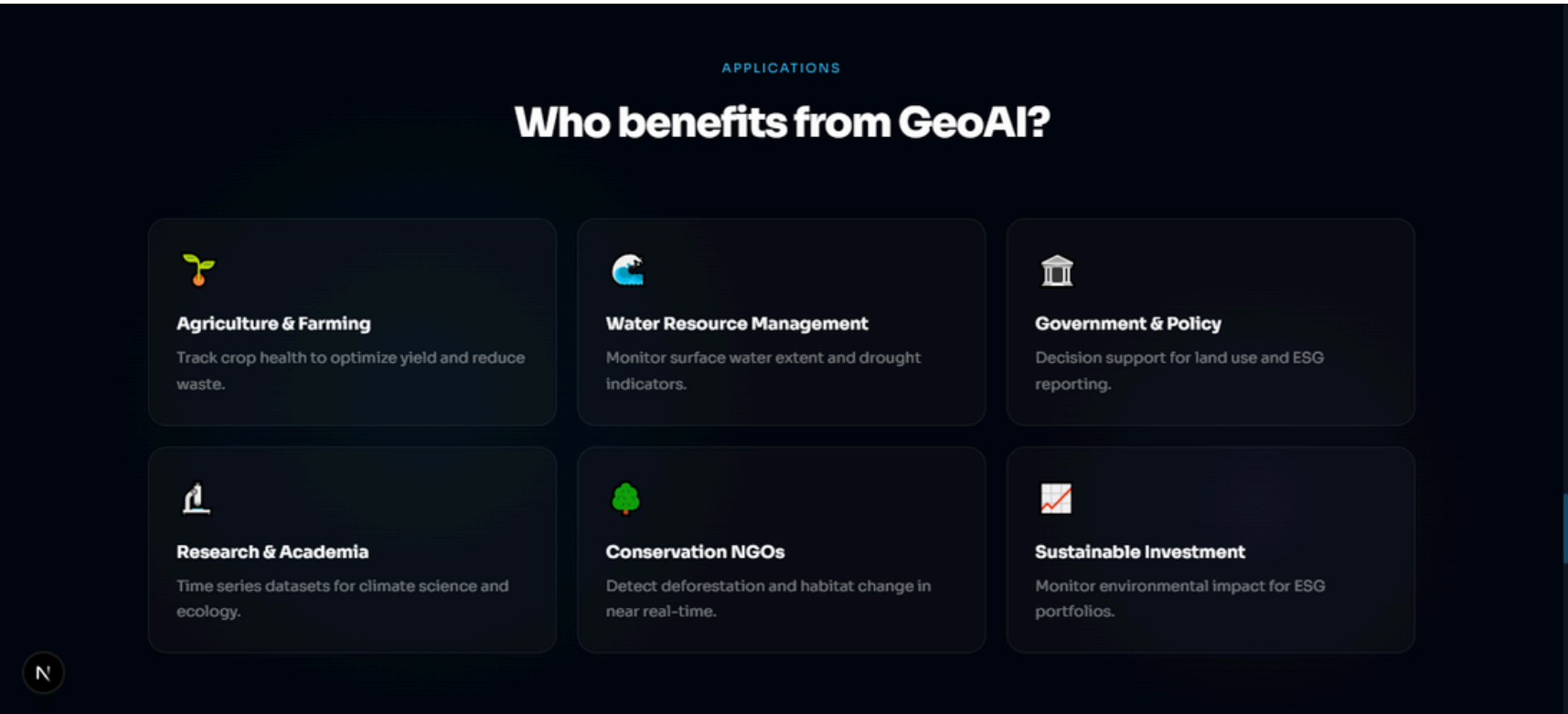
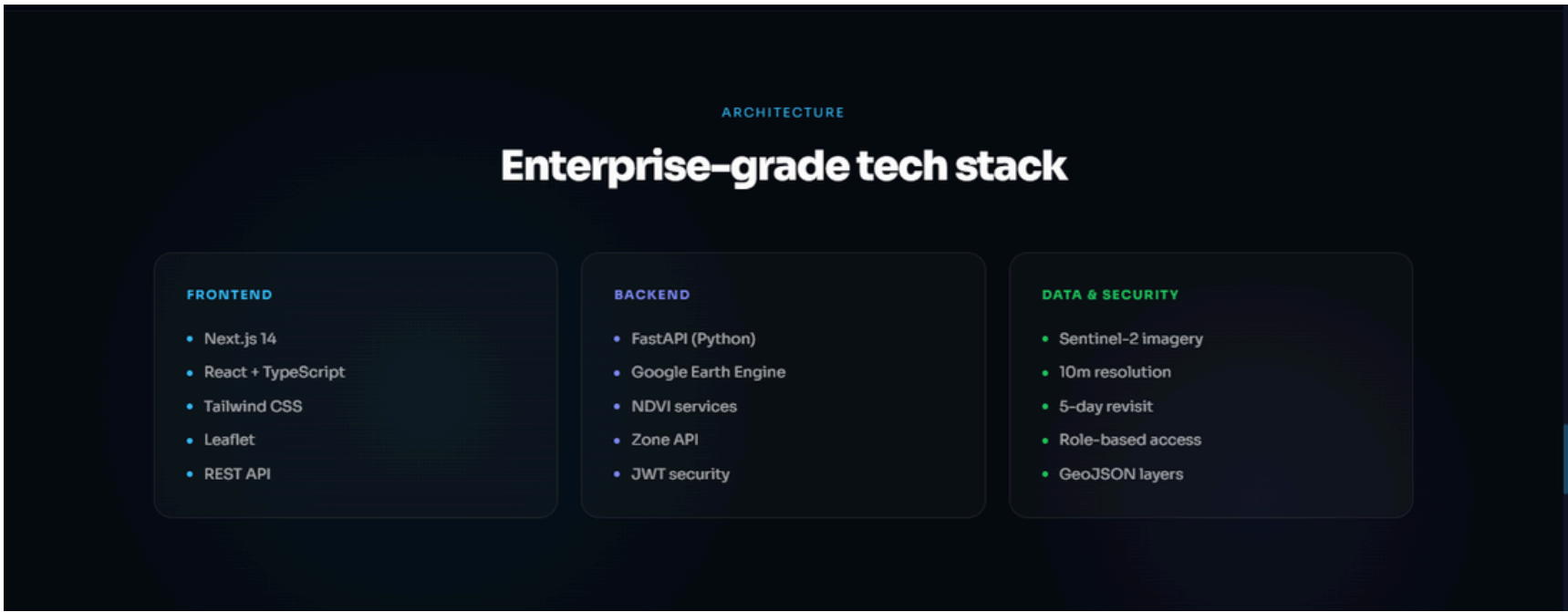
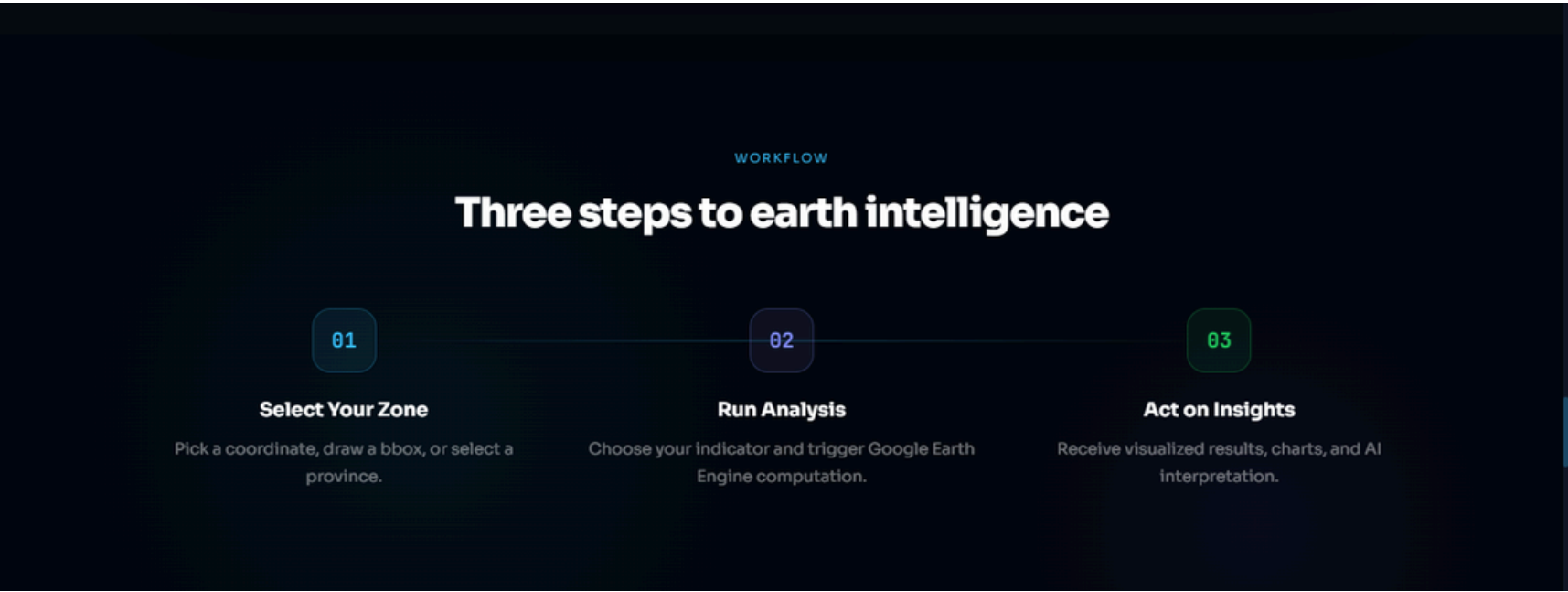
## See the Platform in Action

Watch a real walkthrough of the platform — from zone selection to AI-powered analysis results.

Watch the Platform Walkthrough  
Click anywhere to play demo video

# Description de la plateforme

## > Landing page



# Description de la plateforme

## > Landing page

**Choose your plan**

Limits aligned with Earth Engine costs: area, period, and number of analyses.

Current Plan : **Démo**

| Free Plan                          | Pro Plan                                | Premium Plan                                          | Contrat (B2B)                                         |
|------------------------------------|-----------------------------------------|-------------------------------------------------------|-------------------------------------------------------|
| <b>Free</b>                        | <b>\$100/week</b>                       | <b>\$1000/month</b>                                   | <b>Sur devis</b>                                      |
| <b>Modules:</b> Land Use           | <b>Modules:</b> GW + Surface + Land Use | <b>Modules:</b> GW + Surface + Land Use               | <b>Modules:</b> GW + Surface + Land Use               |
| <b>Zones:</b> point                | <b>Zones:</b> point, bbox, province     | <b>Zones:</b> point, bbox, province, region, national | <b>Zones:</b> point, bbox, province, region, national |
| <b>Max BBox:</b> 0 km <sup>2</sup> | <b>Max BBox:</b> Unlimited              | <b>Max BBox:</b> Unlimited                            | <b>Max BBox:</b> Unlimited                            |
| <b>Max Period:</b> 6 months        | <b>Max Period:</b> 36 months            | <b>Max Period:</b> Unlimited                          | <b>Max Period:</b> Unlimited                          |
| <b>Analyses / month:</b> 10        | <b>Analyses / month:</b> 100            | <b>Analyses / month:</b> 500                          | <b>Analyses / month:</b> Unlimited                    |
| <b>PDF Export:</b> —               | <b>PDF Export:</b> ✓                    | <b>PDF Export:</b> ✓                                  | <b>PDF Export:</b> ✓                                  |
| Upgrade to this plan               | Upgrade to this plan                    | Upgrade to this plan                                  | Upgrade to this plan                                  |

N

# Description de la plateforme

## > Authentification

### HydroSight

Créez votre compte

NOM COMPLET (OPTIONNEL)

EMAIL

MOT DE PASSE

CONFIRMER

**Créer le compte**

Déjà inscrit ? [Se connecter](#)

### HydroSight

Connectez-vous à votre espace

EMAIL

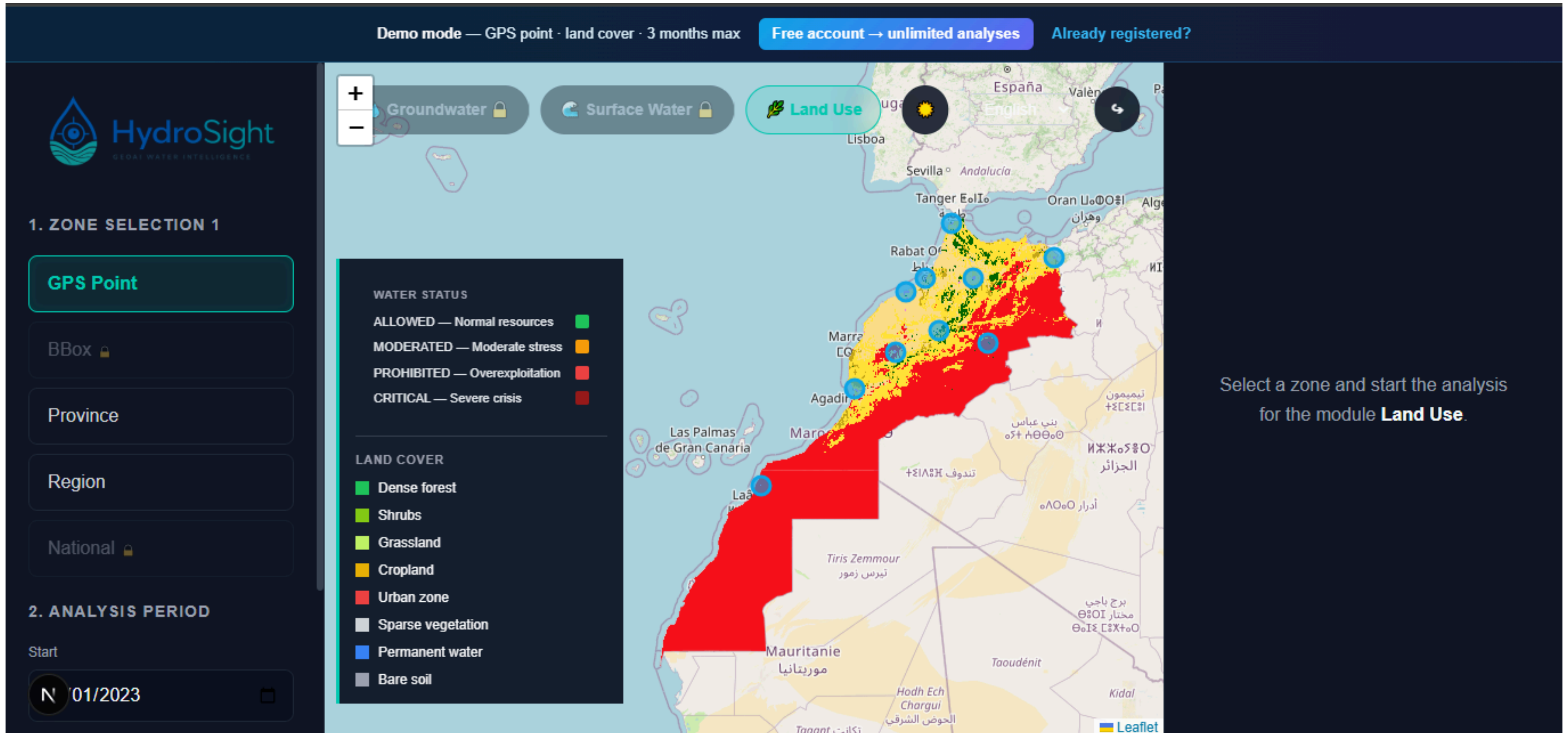
MOT DE PASSE

**Se connecter**

Pas encore de compte ? [Créer un compte](#)

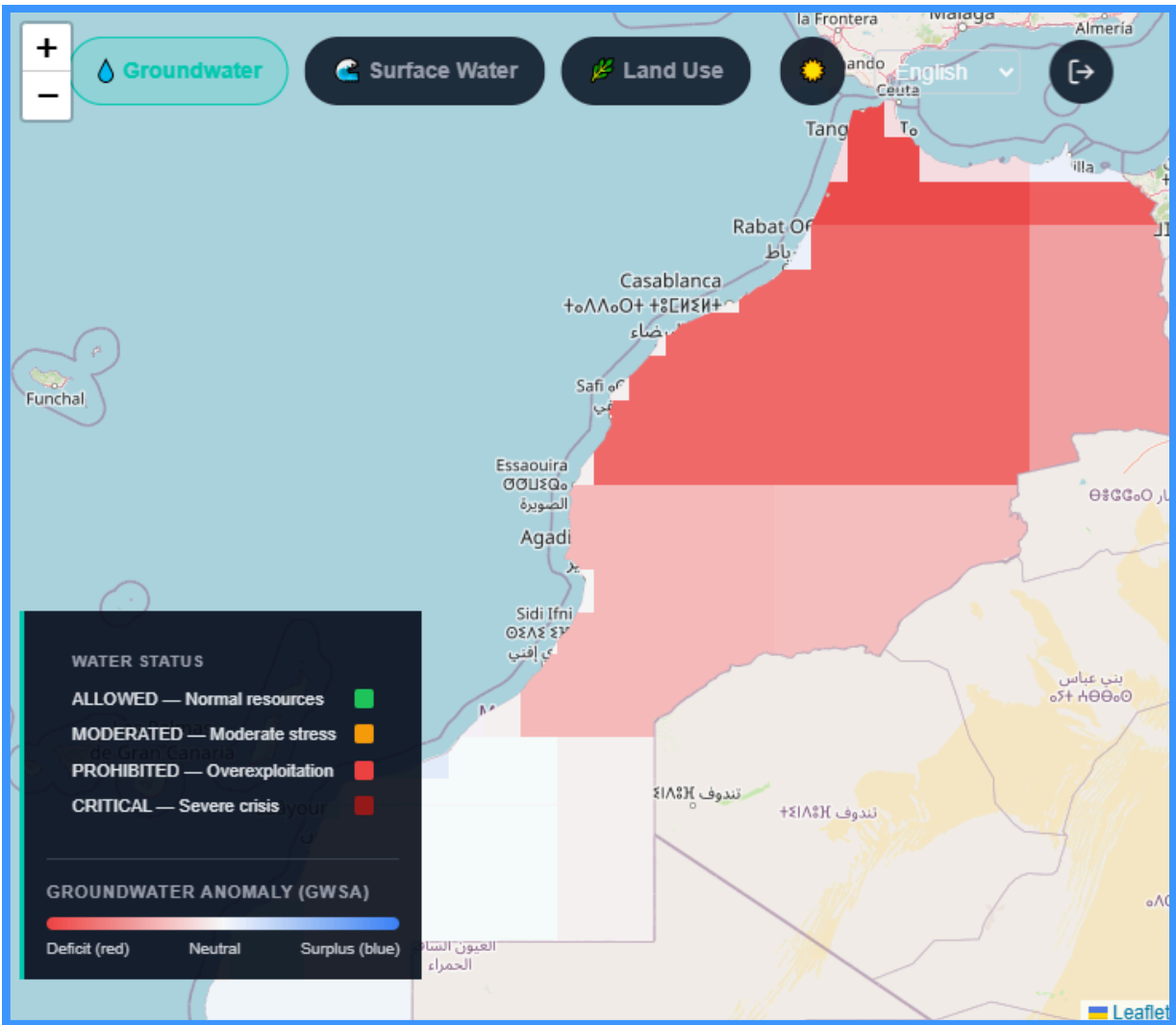
# Description de la plateforme

## ➤ Dashboard



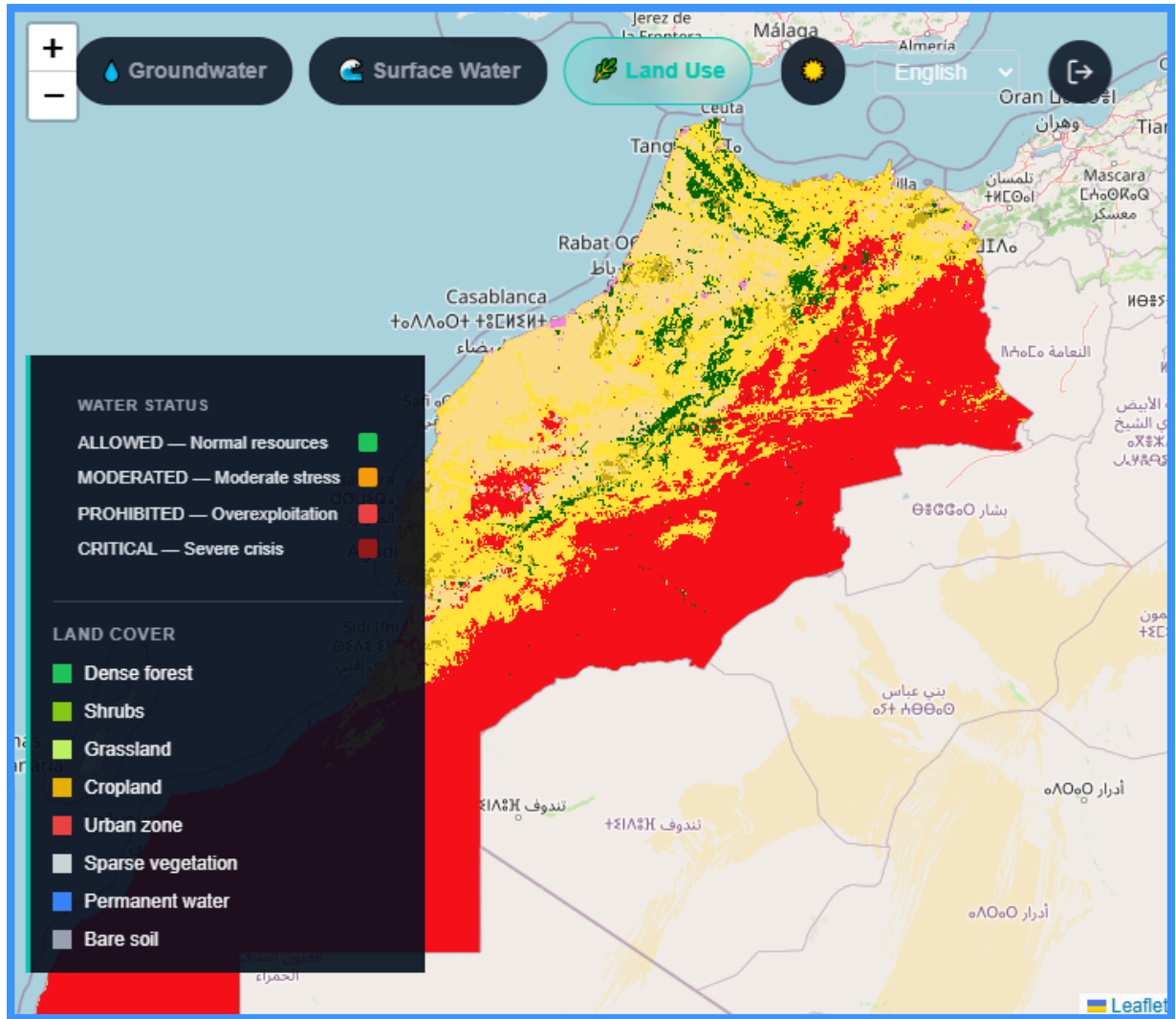
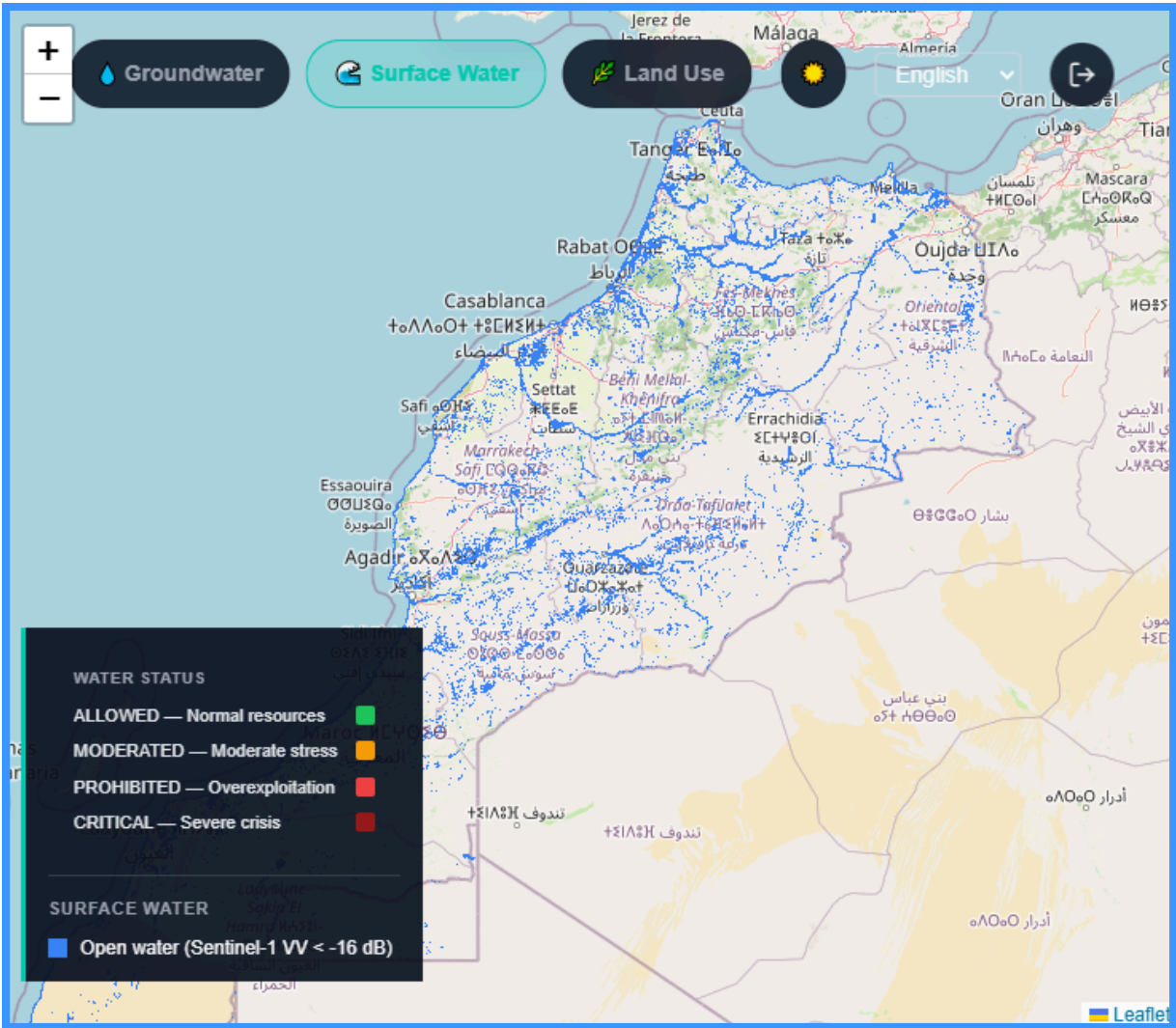
# Description de la plateforme

## Modules



Ground water

Surface water



Land use

# Description de la plateforme

## > paramètres de l'analyse



**PLAN PREMIUM**  
Analyses : 0 / 500

**1. ZONE SELECTION 1**

**GPS Point**

BBox

Province

Region

National

**ACTIVE ZONE**  
**Point (32.7688, -5.1416)**

**2. ANALYSIS PERIOD**

Start  
01/01/2023

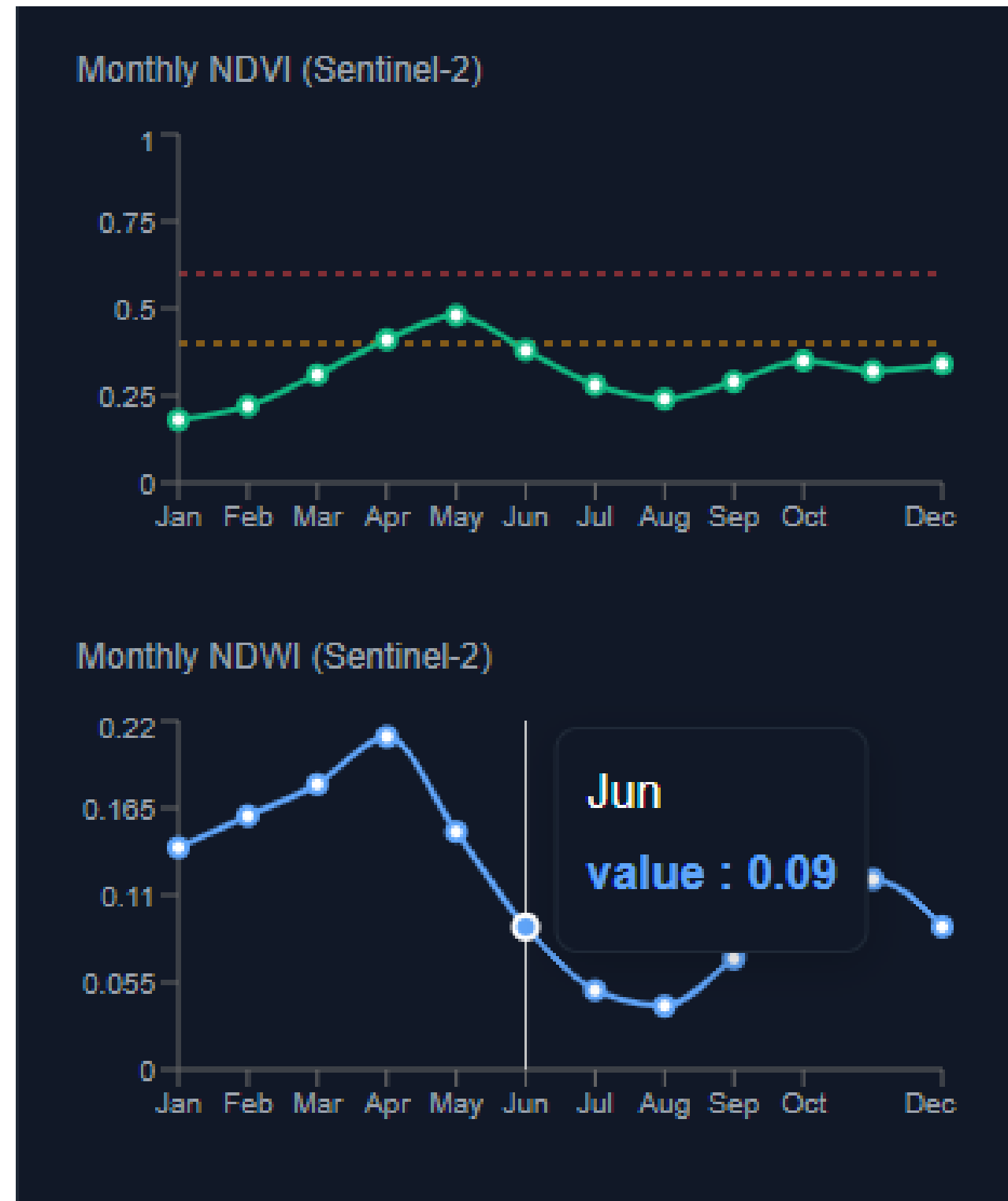
End  
31/12/2023

Duration: 12 months

**N' RUN ANALYSIS**

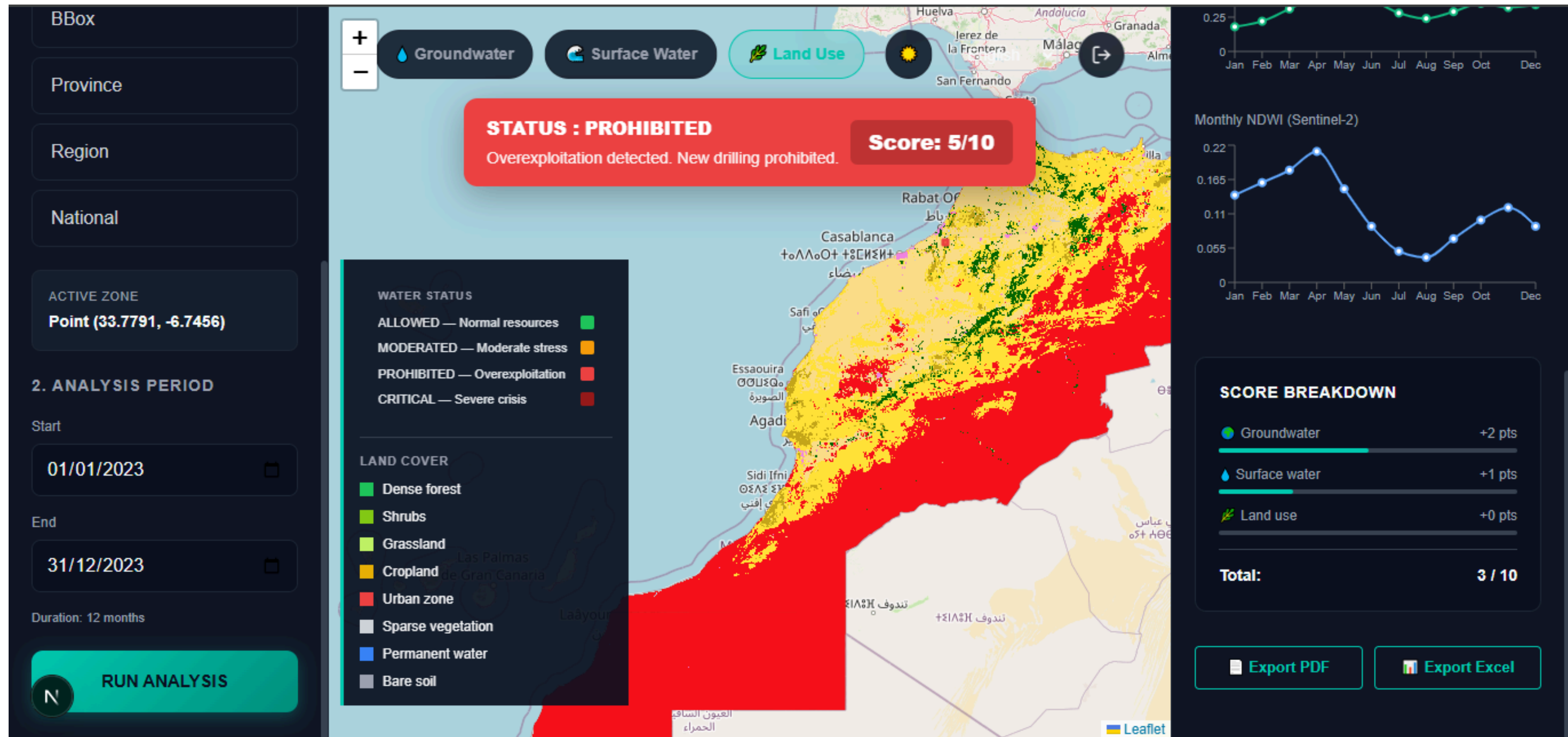
# Description de la plateforme

## ➤ Résultats de l'analyse



# Description de la plateforme

## ➤ Résultats de l'analyse



# Description de la plateforme

➤ PDF exporté

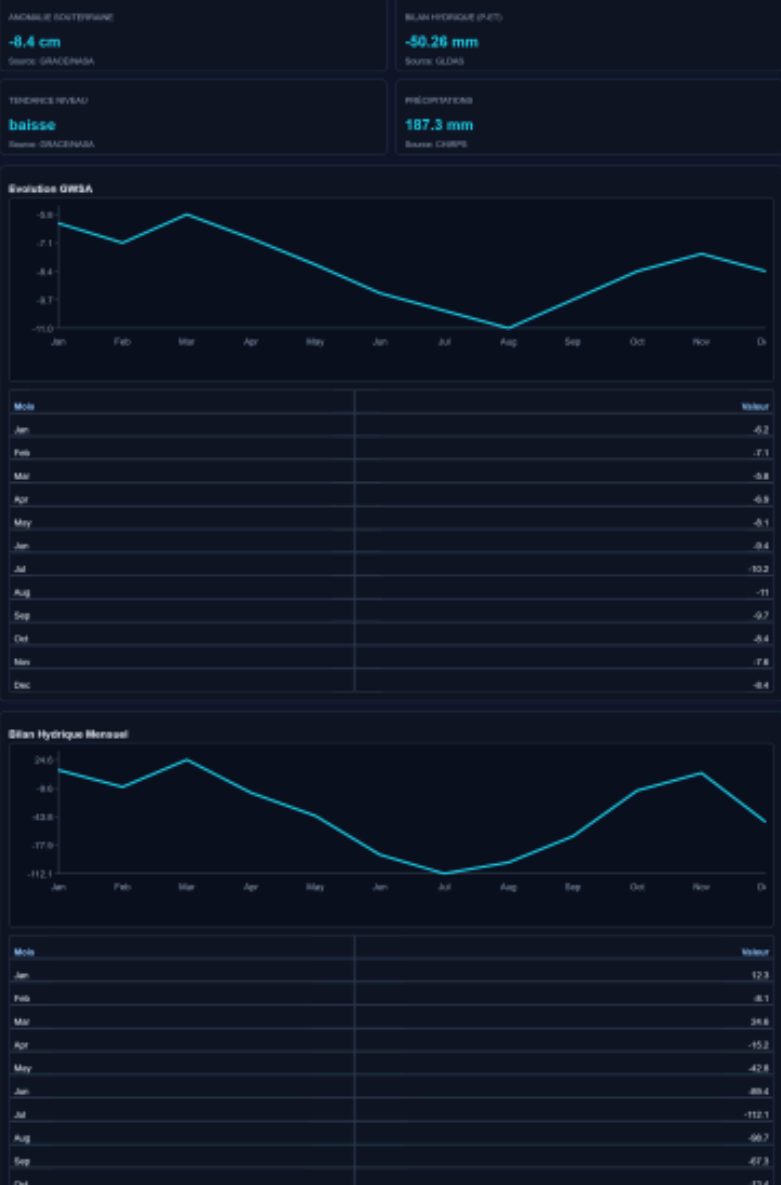
## Rapport HydroSight GeoAI - Complet

Date d'export: 21/06/2026 18:00:23  
Période: 2023-01-01 -> 2023-12-31  
Statut: MODERATED  
Score: 3/10

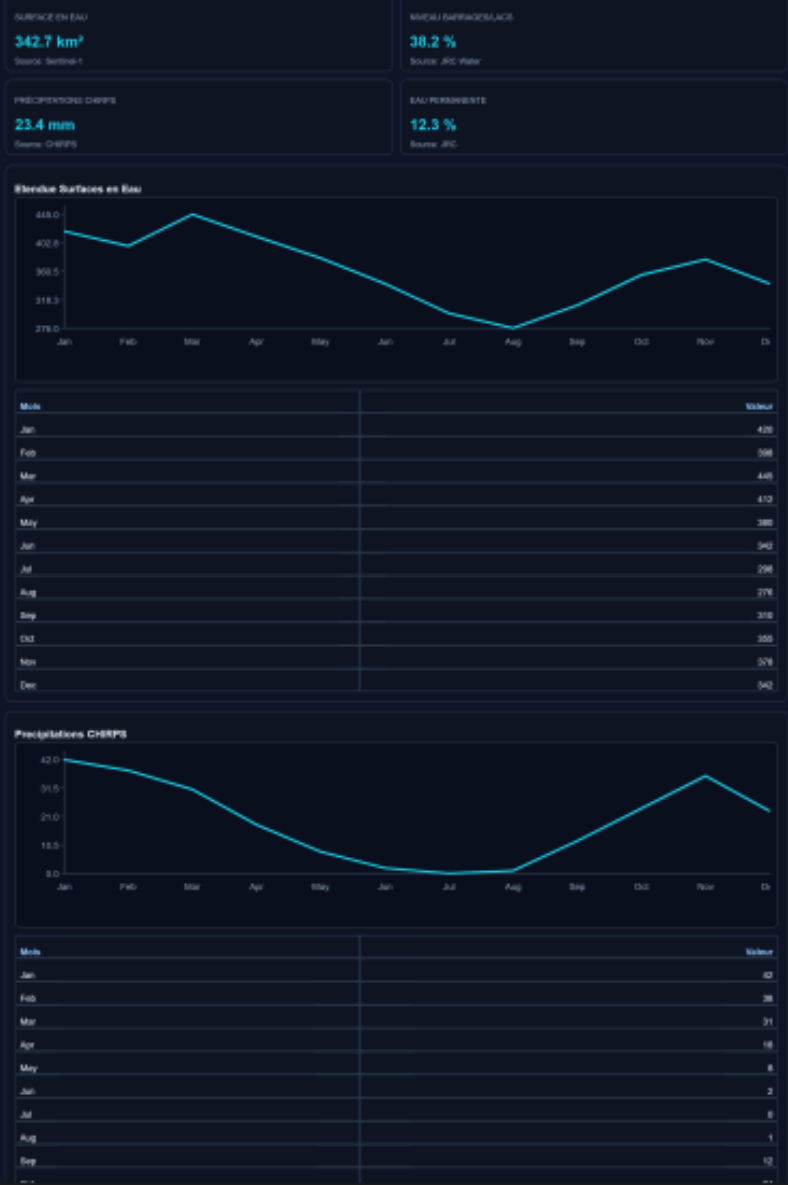
Zone sélectionnée

```
{
  "granularite": "point",
  "lat": 33.779147331286474,
  "lon": -6.745685468750001,
  "label": "Point (33.7791, -6.7456)"
}
```

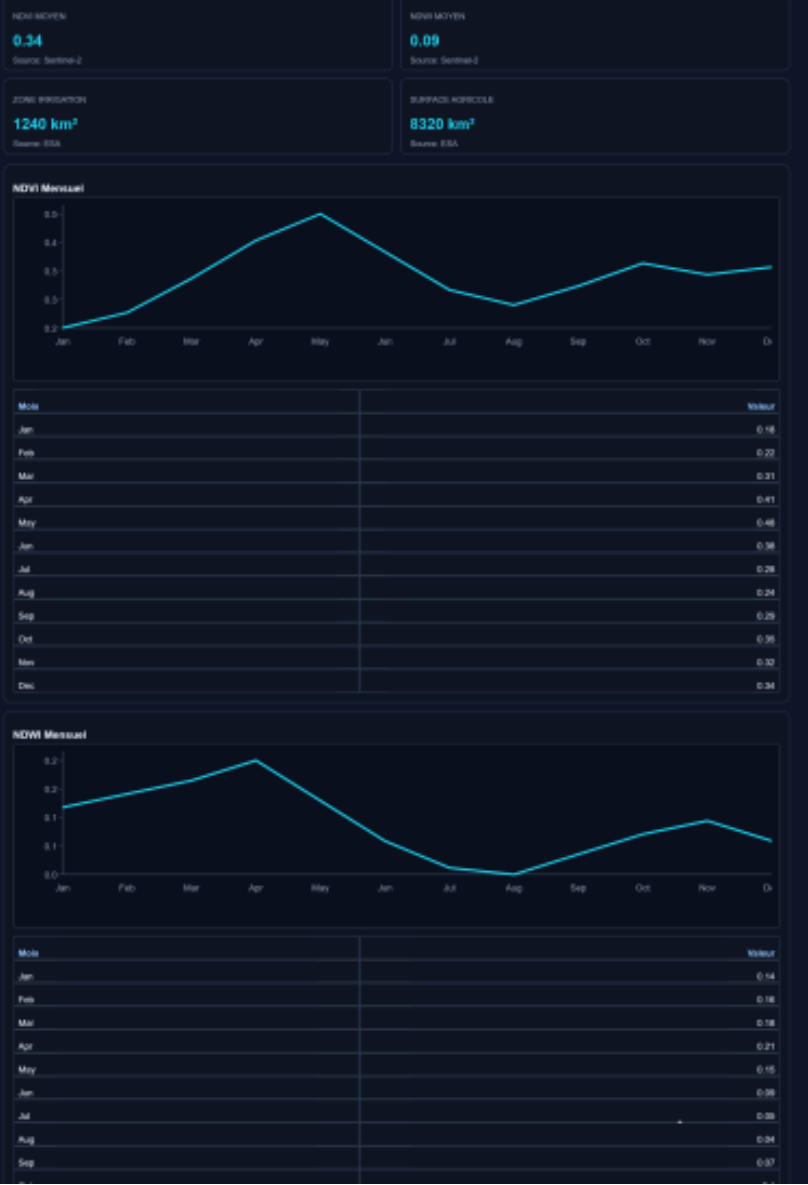
### Module 1 - Eaux Souterraines (GW)



### Module 2 - Eaux de Surface (SW)



### Module 3 - Occupation du Sol (LU)



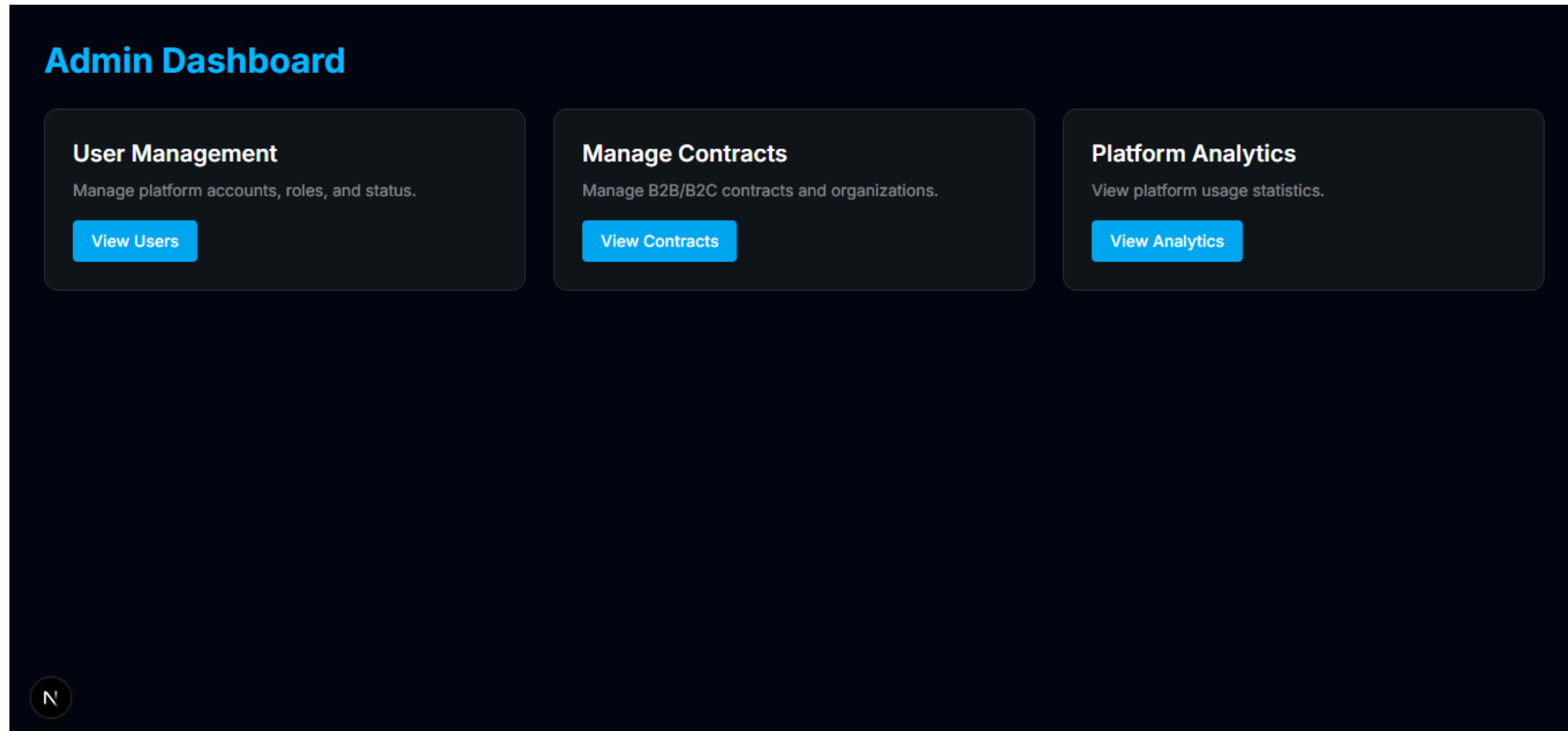
# Description de la plateforme

## > Excel exporté

| A1 |              | generatedAt |            |           |       |         |         |         |            |                                                                                                   |   |   |   |   |   |   |
|----|--------------|-------------|------------|-----------|-------|---------|---------|---------|------------|---------------------------------------------------------------------------------------------------|---|---|---|---|---|---|
|    | A            | B           | C          | D         | E     | F       | G       | H       | I          | J                                                                                                 | K | L | M | N | O | P |
| 1  | generatedAt  | dateDebut   | dateFin    | status    | score | scoreGW | scoreSW | scoreLU | scoreTotal | zoneSelection                                                                                     |   |   |   |   |   |   |
| 2  | 21/06/2026 1 | 2023-01-01  | 2023-12-31 | MODERATED | 3     | 2       | 1       | 0       | 3          | { "granularite": "point", "lat": 33.779147331286474, "lon": -6.745605468750001, "label": "Point ( |   |   |   |   |   |   |

# Description de la plateforme

## > Espace admin



# Description de la plateforme

## > User management

[← Back](#) **User Management**

| EMAIL                    | ROLE    | PLAN      | JOINED     |
|--------------------------|---------|-----------|------------|
| admin@hydrosight.com     | admin ▼ | premium ▼ | 13/06/2026 |
| anejarahalima@gmail.com  | user ▼  | premium ▼ | 31/05/2026 |
| soumiaamerkad@gmail.com  | user ▼  | free ▼    | 12/05/2026 |
| soumayaamerkad@gmail.com | user ▼  | pro ▼     | 06/05/2026 |

# Description de la plateforme

## > Contracts

[← Back](#) **Contracts** [+ New Contract](#)

User ID \*

uuid of the user

Type

B2B

Plan

Pro

Starts At

jj/mm/aaaa

Expires At

jj/mm/aaaa

Notes

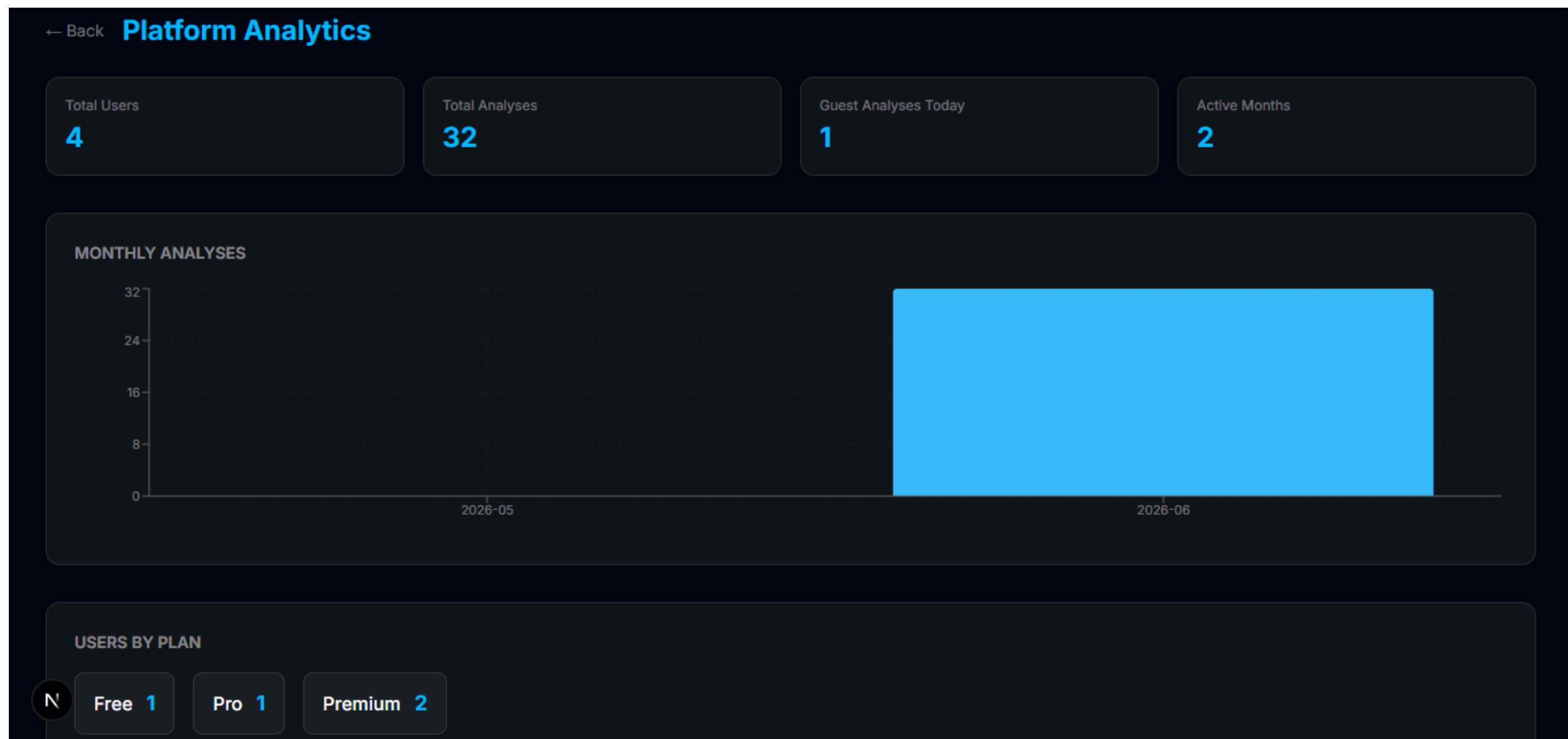
Create Contract

Cancel

| USER ID     | TYPE | PLAN | STATUS  | EXPIRES    | ACTIONS |
|-------------|------|------|---------|------------|---------|
| 3d9c783a... | B2B  | Pro  | pending | 12/07/2026 | pending |

# Description de la plateforme

## > Platform analytics





# Conclusion



# Merci pour votre attention

